

Interest rate derivatives

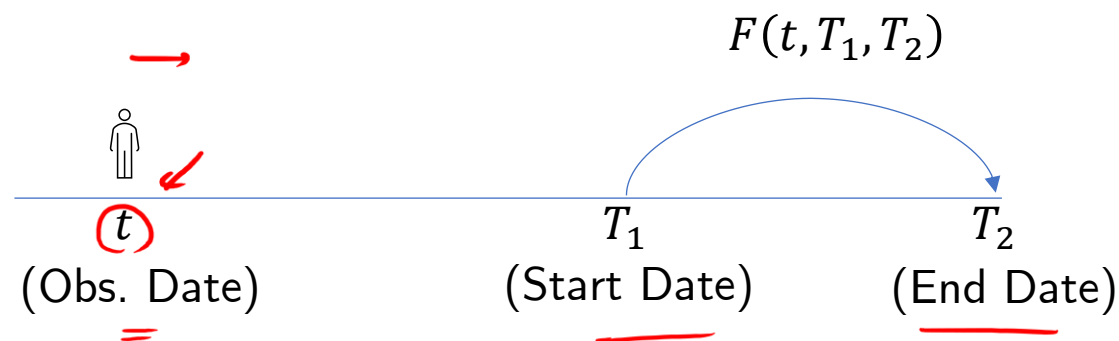
1. Linear Derivatives

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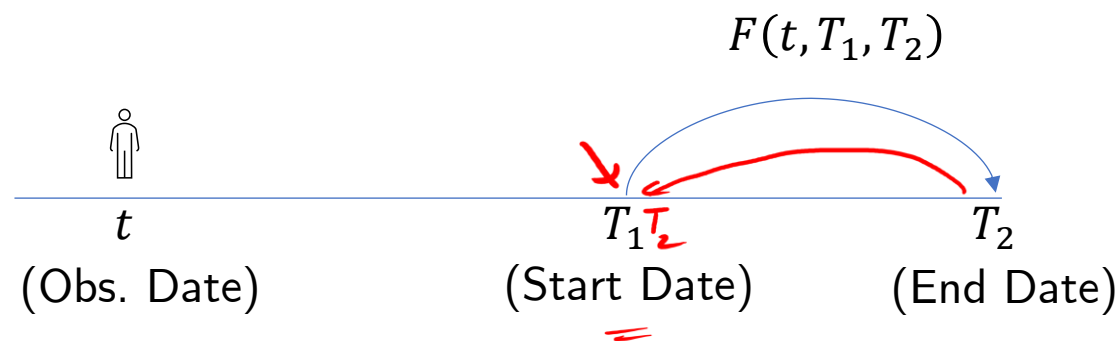
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Rate



- An interest rate object is characterized by 3-time instances (t, T_1, T_2) where $t \leq T_1 < T_2$
- Obs. Date (when the rate is available)
 - Start Date (starting date of the rate period)
 - End Date (ending date of the rate period)

Forward Rate



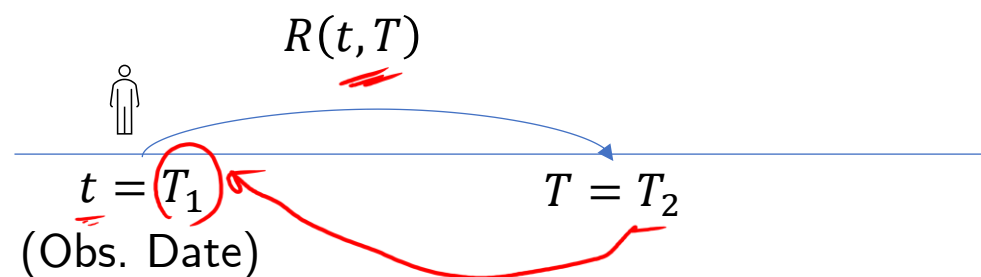
► When $t < T_1$, we call it a **Forward Rate**. Think of this as booking a hotel in advance. Say, you are trying to book a hotel for 2 nights from 31/12/2024(T_1) to 2/1/2025(T_2) on 20/10/2024(t). The rate available on 20/10/2024(t) is given as @200/night. Similarly, an interest rate can be locked in advance for a period in future and is always quoted per annum basis.

► if we bring $T_2 \rightarrow T_1$, the rate applies over an **infinitesimal period**, and therefore represents **instantaneous forward rate**. It's characterized by 2-time instances (t, T) . We will denote it as $f(t, T)$

$$\uparrow \quad \quad \quad \textcircled{t} \quad \quad \quad T \quad \quad \quad T + \delta \uparrow$$

Spot Rate

$F(t, T_1, T_2)$
 $f(t, T)$
 $R(t, T) \rightarrow$
 $\uparrow$



► A spot rate object is characterized by 2-time instances ($\underline{t = T_1}, \underline{T = T_2}$). The start of the period (T_1) coincides with the Obs. Date (t). This is called as spot rate, and we will denote it as $\underline{R(t, T)}$. Think of this as you walking into a hotel with no advance booking and agreeing to stay for 2 nights. The rate you will be charged is the spot rate.

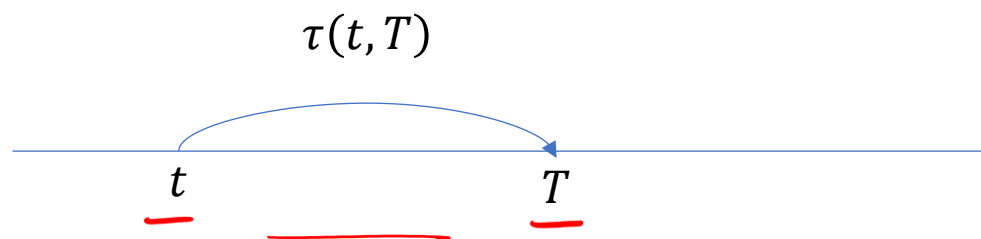
► If we bring $T_2 \rightarrow T_1$, the rate applies over an infinitesimal period, and therefore represents instantaneous spot rate. It's also popularly called as short rate.

► A short rate is characterized by one time instance ($\underline{t}$), so we will denote it as $\underline{r(t)}$

$(t, t, t+\delta)$

Year Fraction : $\tau(t, T)$

1/year × years



► To calculate the interest payment over a period which involves two dates (D_1, D_2), we need a measure that calculates the #years between the two dates. We have a few popular conventions

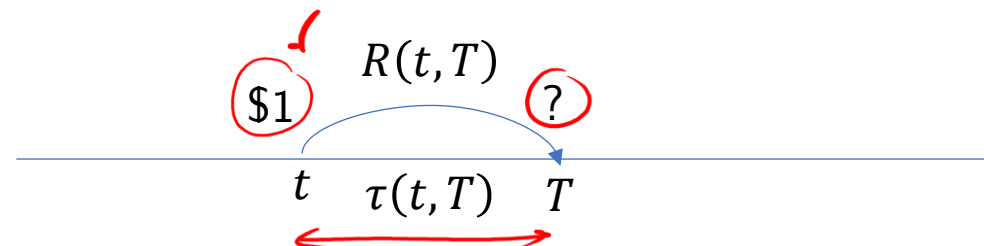
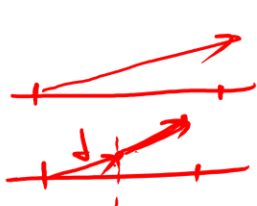
► Say $D_1 = \{d_1, m_1, y_1\}$, $D_2 = \{d_2, m_2, y_2\}$

• $\text{Act}/365 = \frac{D_2 - D_1}{365}$

• $\text{Act}/360 = \frac{D_2 - D_1}{360}$

• $30/360 = \frac{\max(30 - d_1, 0) + \min(d_2, 30) + 360(y_2 - y_1) + 30(m_2 - m_1 - 1)}{360}$

Compounding Frequency

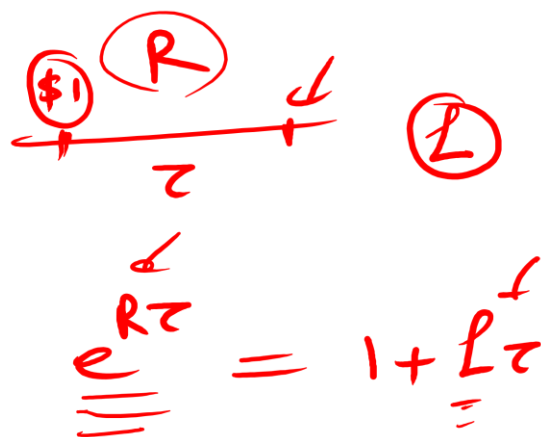


► To calculate interest over a period, we also need the compounding frequency which basically breaks the period into sub-periods and compounds the interest after each sub-period. Let's assume we invest \$1 at the start of the period (t). Let's calculate the money we will have at the end of the period (T)

- Continuous Compounding : $e^{R(t,T)\tau(t,T)}$
- Simple Compounding : $(1 + L(t, T)\tau(t, T))$
- Annual Compounding : $(1 + Y(t, T))^{\tau(t, T)}$
- K-times per year compounding : $\left(1 + \left(\frac{Y_k}{K}\right)\right)^{K\tau(t, T)}$

Conversion - different compounding types

- We can easily convert one type of rate to another. For example, if we want to convert a continuously compounded rate (R) to a simply compounded rate (L) or vice versa, then we need to do the following



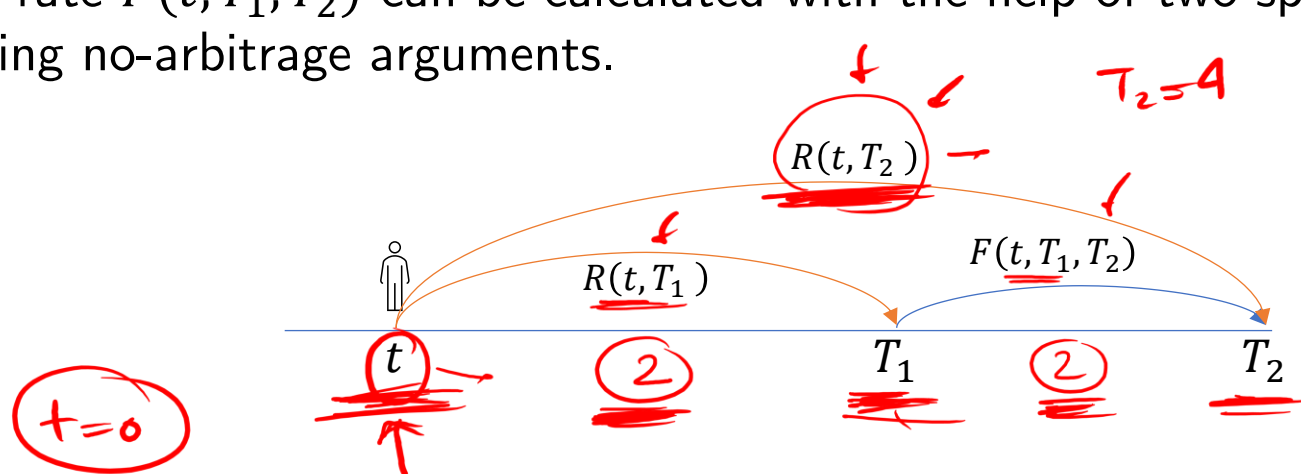
$$e^{R\tau} = 1 + L\tau$$

$$e^{R\tau} = 1 + L\tau$$

$$L = \frac{1}{\tau}(e^{R\tau} - 1)$$

Conversion – Spot vs Forward

- ▶ A forward rate $F(t, T_1, T_2)$ can be calculated with the help of two spot rates $R(t, T_1)$ and $R(t, T_2)$ using no-arbitrage arguments.



$$F(t, T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left(\frac{e^{R(t, T_2)\tau(t, T_2)}}{e^{R(t, T_1)\tau(t, T_1)}} - 1 \right)$$

$$= \frac{1}{\tau(T_1, T_2)} \left(\frac{P(t, T_1)}{P(t, T_2)} - 1 \right)$$

(Continuous compounding)

$$e^{R(t, T_2)\tau(t, T_2)} = e^{R(t, T_1)\tau(t, T_1)} e^{F(t, T_1, T_2)\tau(T_1, T_2)}$$

$$F(t, T_1, T_2) = \frac{R(t, T_2)\tau(t, T_2) - R(t, T_1)\tau(t, T_1)}{\tau(T_1, T_2)}$$

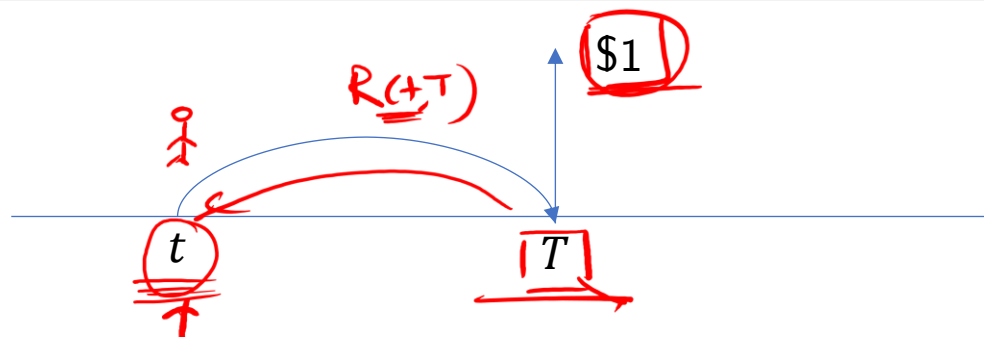
$\frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$

(Simple compounding)

$$e^{R(t, T_2)\tau(t, T_2)} = e^{R(t, T_1)\tau(t, T_1)} (1 + F(t, T_1, T_2)\tau(T_1, T_2))$$

$$F(t, T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left(\frac{P(t, T_1)}{P(t, T_2)} - 1 \right)$$

Instrument #1 – Zero Coupon Bond(ZCB)



- A Zero-coupon Bond(ZCB) pays \$1 at maturity (T). The valuation of the ZCB at time (t) is linked to the spot rate and vice versa

(continuous compounding)

$$P(t, T) = e^{-R(t, T)\tau(t, T)}$$

valuation date $\nwarrow$
 $\nearrow$ maturity

Continuously compounded

$$R(t, T) = -\frac{1}{\tau(t, T)} \ln P(t, T)$$

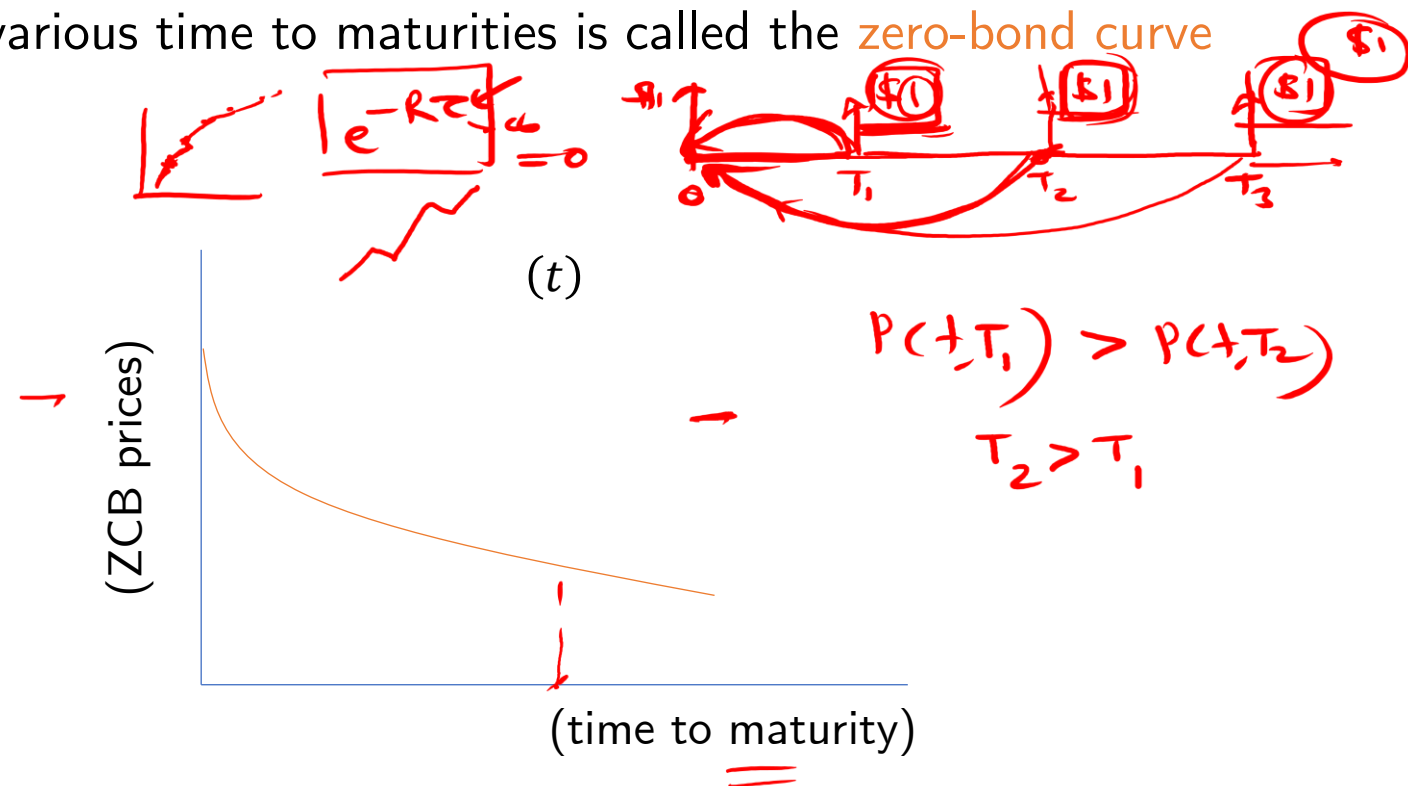
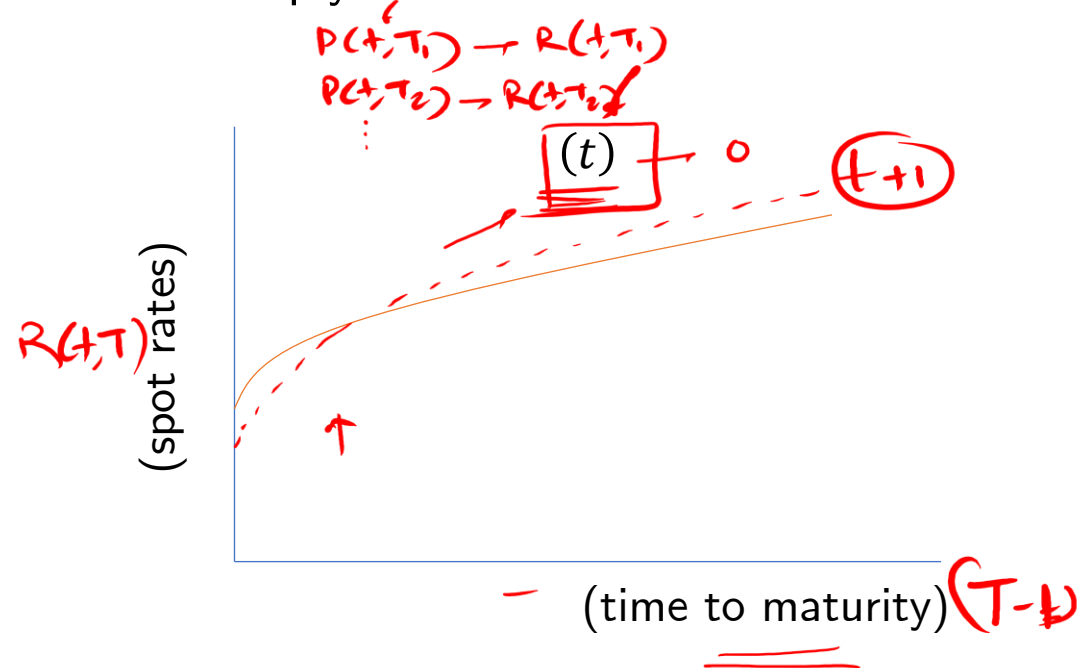
(simple compounding)

$$P(t, T) = \frac{1}{1 + L(t, T)\tau(t, T)}$$

$$L(t, T) = \frac{1}{\tau(t, T)} \left(\frac{1}{P(t, T)} - 1 \right)$$

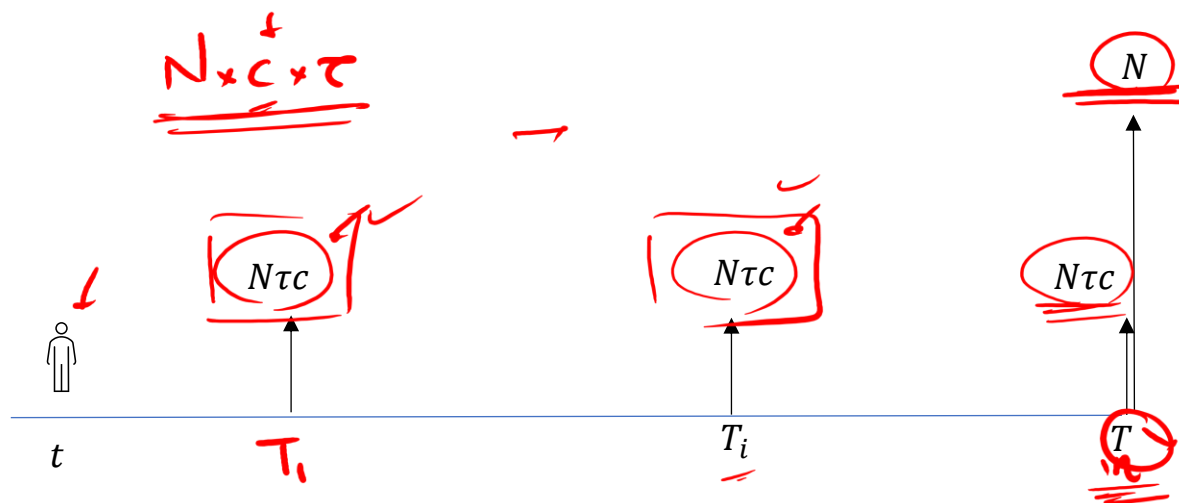
Zero Yield Curve and Zero Bond Curve

- ▶ On any observation date (t), we observe a series of zero-coupon bonds for various maturities. We can back out the spot rates from ZCB prices. The plot of the spot rates vs time to maturity ($T - t$) is called the **zero-coupon yield curve** (also called **term structure** of rates)
- ▶ Similarly, the plot of ZCB prices against various time to maturities is called the **zero-bond curve** or simply **discount curve**.



Instrument #2 – Fixed Coupon Bond(FCB)

- ▶ A Fixed Coupon bond pays a fixed coupon rate periodically until maturity. At maturity it also pays the principal or face value.
- ▶ The Valuation of an FCB can be seen as a linear combination of zeros where the coupon and the principal act as the scalars and zeros act as the discount factors.



$$FCB(t, \mathcal{T}, N, \tau, c, \omega)$$

t : Valuation date

$\mathcal{T}$: coupon payment schedule

N : Notional

τ : settlement frequency

c : coupon rate

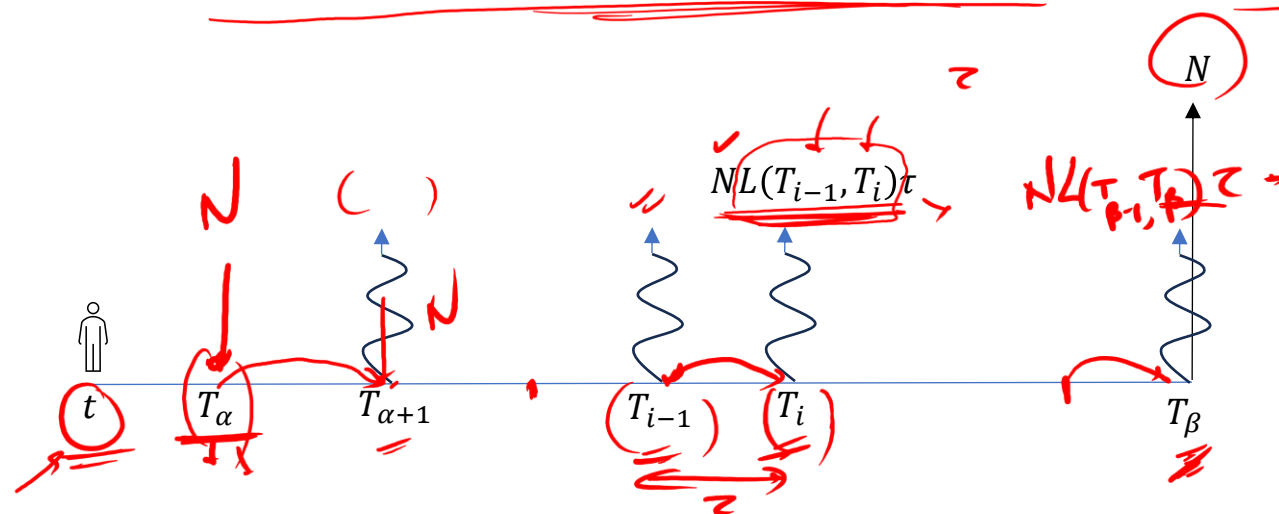
ω : Position (+1 : long, -1 : short)

yield curve

$$FCB(t, \mathcal{T}, N, \tau, c, \omega) = N\tau c \sum_{i=1}^n P(t, T_i) + NP(t, T_n)$$

Instrument #3 – Floating rate Note(FRN)

- ▶ A Floating Rate Note (FRN) pays a floating coupon periodically until maturity. At maturity it also pays the principal or face value. The floating payments become known on the reset date prior to the payment.
- ▶ The FRN always is valued at par (N) on any reset date.



$$FRN(t, T_\alpha, T_\beta, N, \tau, c, \omega)$$

t : Valuation date

T_α : First Reset Date

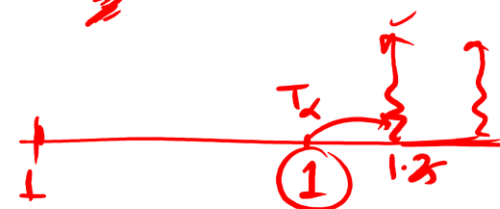
T_β : Last Reset Date

N : Notional

τ : settlement frequency

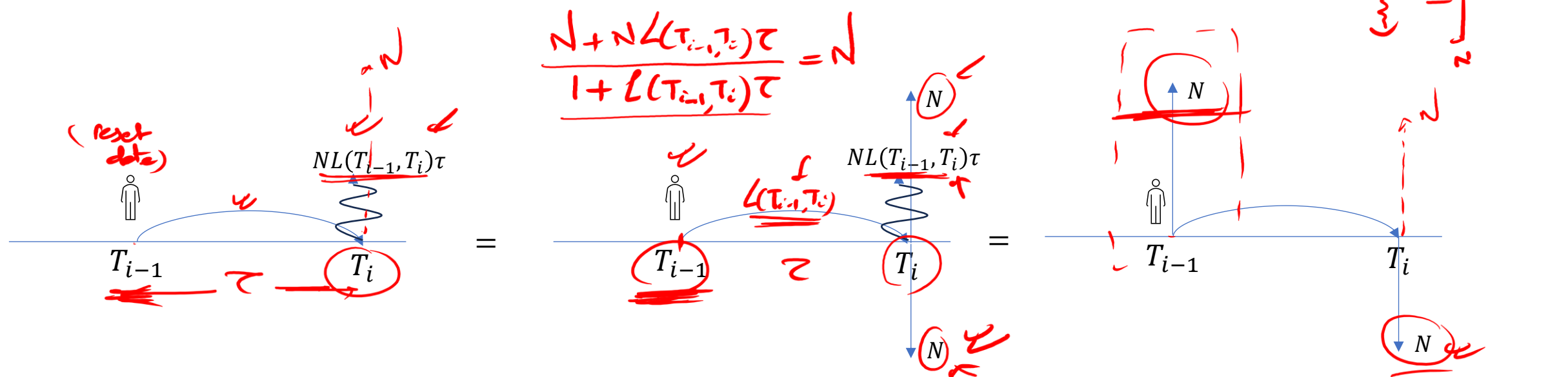
ω : Position (+1 : long, -1 : short)

$$FRN(t, T_\alpha, T_\beta, N, \tau, c, \omega) = N(1 + \tau c)P(t, T_{next})$$



Why Floater discounts to par ?

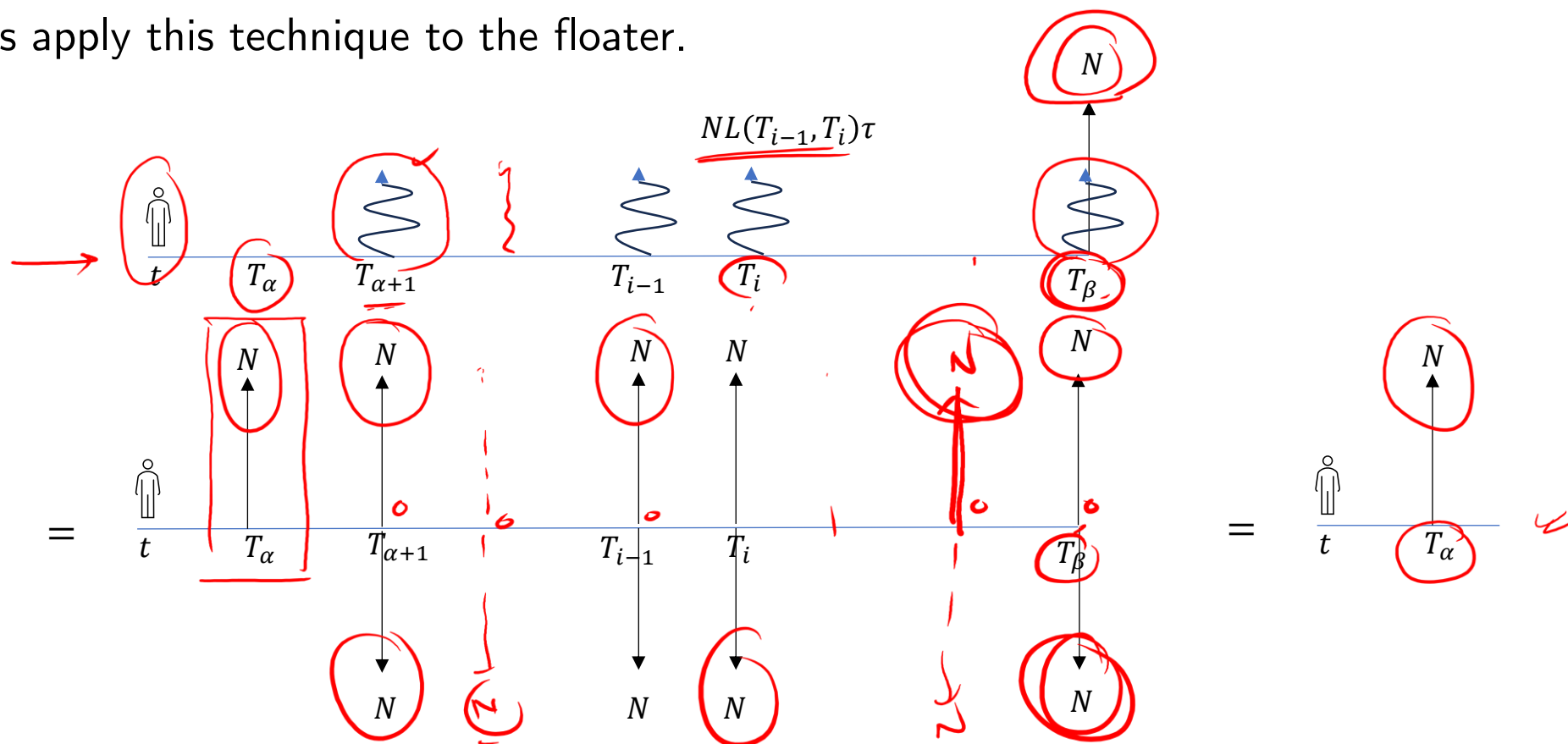
- ▶ To understand why floater discounts to par, consider a floating interest payment which just got fixed on the reset date where we want to value it. For the period (T_{i-1}, T_i) , the simple compounded rate $L(T_{i-1}, T_i)$ becomes visible at T_{i-1}



- ▶ We can add and subtract a payment N at T_i . The inflow $N(1 + L(T_{i-1}, T_i)\tau)$ is discounted to N at T_{i-1} . So essentially a floating payment can be seen as getting N one period earlier and paying N one period later.

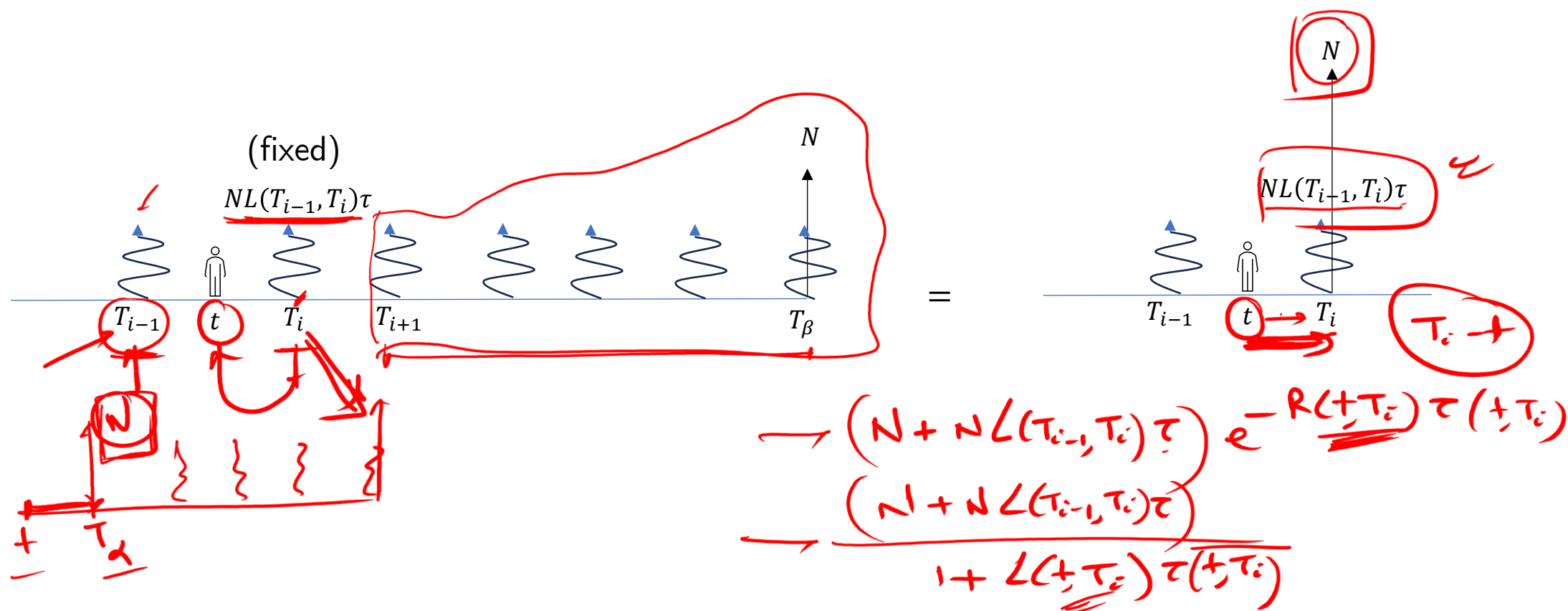
Why Floater discounts to par ?

► Now, let's apply this technique to the floater.



Floater in between two reset dates

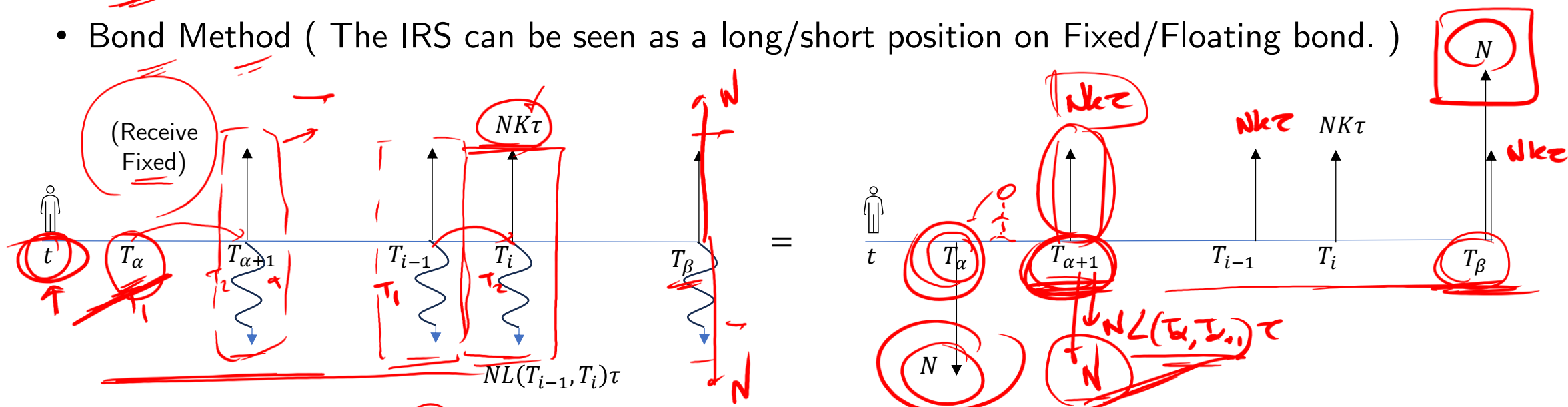
- If the valuation date is between two reset dates, then the first floating coupon would have already been fixed on the previous reset date. All remaining floating payments including principal at the end will discount to par.



Instrument #5 – Interest rate swap (IRS)

▶ A vanilla IRS involves swapping fixed rate payments and floating rate payments. It can be valued using one of the following two methods

- FRA method (The IRS can be decomposed into a series of FRAs)
- Bond Method (The IRS can be seen as a long/short position on Fixed/Floating bond.)

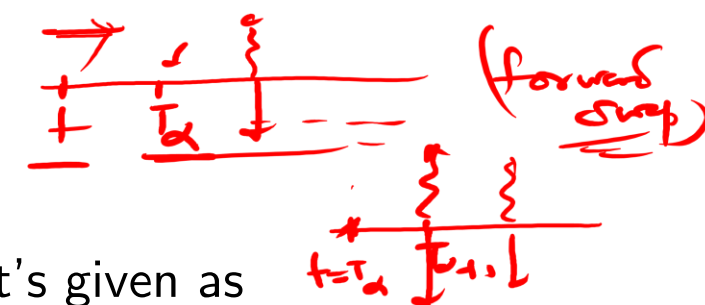


$$IRS(t, T_\alpha, T_\beta, N, \tau) = N\tau K \sum_{i=\alpha+1}^{\beta} P(t, T_i) + \underline{NP(t, T_\beta)} - \underline{NP(t, T_\alpha)}$$

Swap rate : $s_{\alpha,\beta}(t)$

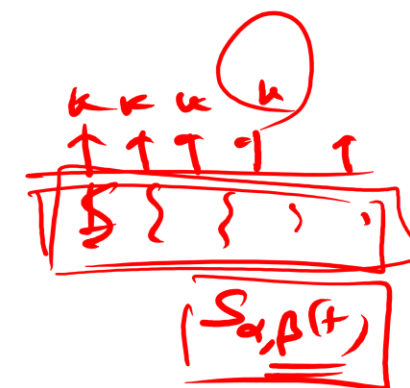
- ▶ The fixed rate that makes the value of IRS = 0 is called the swap rate. As time progresses the valuation of floater changes and therefore the fixed rate that will make the fixed bond value = floating bond value will also evolve. We will denote it by $s_{\alpha,\beta}(t)$

$$s_{\alpha,\beta}(t) = \frac{P(t, T_\alpha) - P(t, T_\beta)}{\tau \sum_{i=\alpha+1}^{\beta} P(t, T_i)}$$

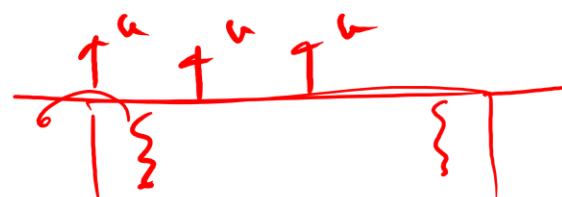


- ▶ At the inception the strike K is set such as the valuation of IRS = 0. It's given as

$$K = s_{\alpha,\beta}(0) = \frac{P(0, T_\alpha) - P(0, T_\beta)}{\tau \sum_{i=\alpha+1}^{\beta} P(0, T_i)}$$



- ▶ The value of IRS alternatively can be given in terms of the swap rate at time t



$$IRS(t, T_\alpha, T_\beta, N, \tau) = N\tau \left(K - s_{\alpha,\beta}(t) \right) \sum_{i=\alpha+1}^{\beta} P(t, T_i)$$